

## Chapter 3

# Trading Short Premium

Options are highly versatile instruments. They can be used to hedge the directional risk of a stock, or they can be used as a source of profits. As alluded to in the example of hurricane insurance, short premium positions can be used to generate small, consistent profits for those willing to accept the tail risk. The mechanics of short premium trading are subtle, but many of the core concepts can be introduced in an intuitive way with some simple gambling analogies. For example, when using options for profit generation (i.e., not risk mitigation), the long-term performance of long and short options can be analogized with slot machines.

- **Buying Options for profit** is like playing the slot machines. Gamblers who play enough times may hit the jackpot and receive a huge payout. However, despite the potential payouts, most players average a loss in the long run because they are taking small losses the majority of the time. Investors who buy options are betting on large, often directional moves in the underlying asset. Those assumptions may be correct and yield significant profits occasionally, but underlying prices ultimately stay within their expected ranges most of the time. This results in small, frequent losses on unused contracts and an average loss over time.
- **Selling Options for profit** is like owning the slot machines. Casino owners have the long-run statistical advantage for every game, an edge particularly high for slots. Owners may occasionally pay out large jackpots, but as long as people play enough and the payouts are manageable, they are compensated for taking on this risk with nearly guaranteed profit in the long term. Similarly, because short options carry tail risk but provide small, consistent profits from implied volatility (IV) overstatement, then they should average a profit in the long run if risk is managed appropriately.

Long premium strategies have a high profit potential but cannot be consistently timed to ensure profit in the long term. This is because outlier underlying moves and IV expansions that benefit long premium positions are strongly linked to external events (such as natural disasters or political conflict), which are relatively difficult to reliably predict. Short premium strategies, on the other hand, profit more often and have the long-term statistical advantage if investors manage risks appropriately.

Similar to the slot machine owner, a short premium trader must reduce the impact of outlier losses to reach a large number of occurrences (trades) and realize the positive long-term averages. This is most effectively done by limiting position size and by adjusting portfolio exposure according to current market conditions. This chapter will, therefore, cover the following broader concepts in volatility trading:

1. Trading in high IV: Identifying favorable conditions for opening short premium trades.
2. Number of occurrences: Reaching the minimum number of trades required to achieve long-term averages.
3. Portfolio allocation and position sizing: Establishing an appropriate level of risk for the given market conditions.
4. Active management and efficient capital allocation: Understanding the benefits of managing trades prior to expiration.

IV plays a crucial role in trading short premium. Remember that IV is a measure of the *sentiment* of uncertainty in the market. It is a proxy for the amount of *fear* among premium buyers (or *excitement*, depending on your personality) and a measure of *opportunity* for premium sellers. When market uncertainty increases, premium prices also increase, and premium sellers receive more compensation for being exposed to large losses. However, IV is also instrumental when managing exposure to extreme losses and establishing appropriate position sizes.

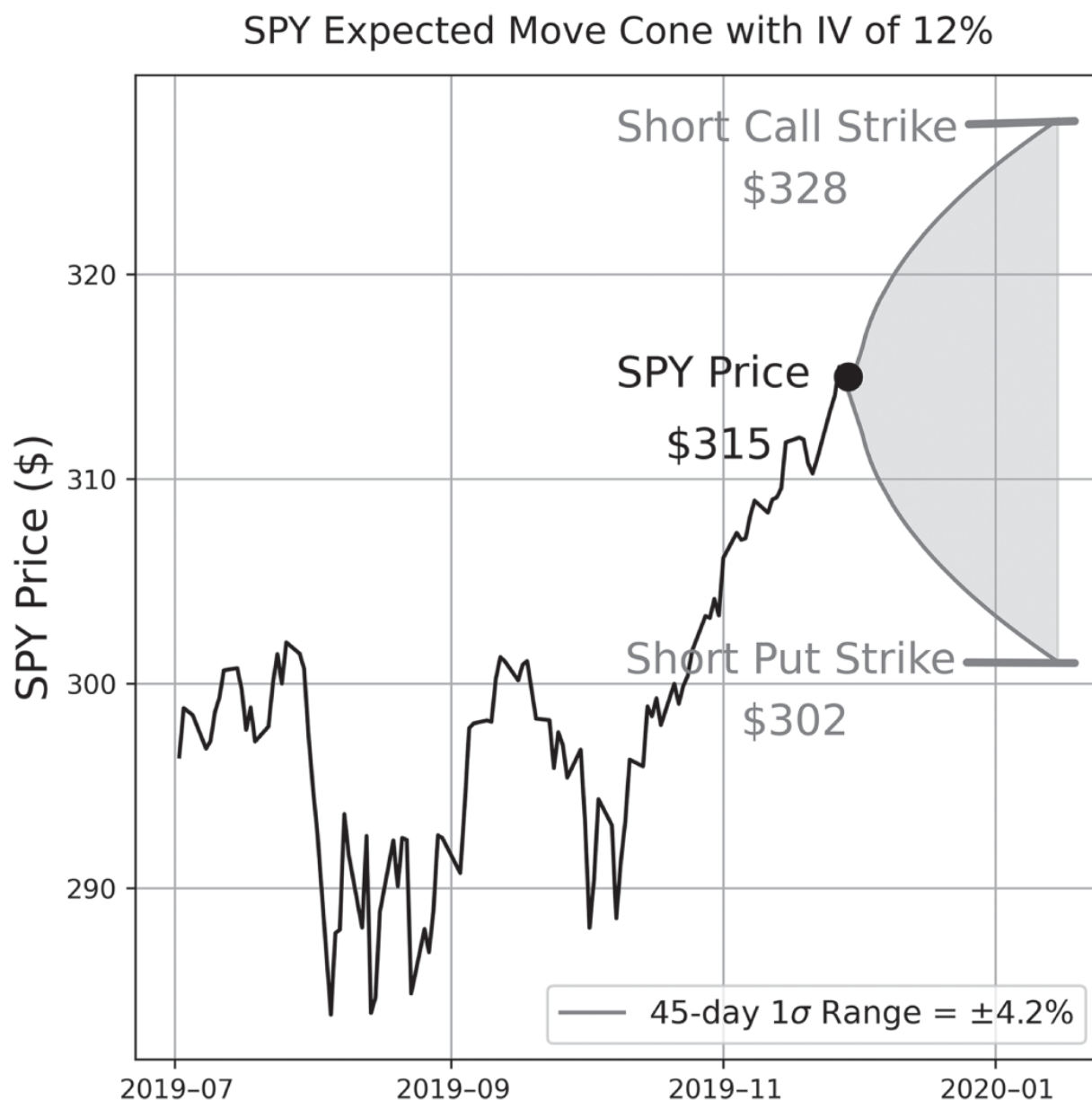
## **Background: A Note on Visualizing Option Risk**

When discussing the risk-reward trade-off of trading short premium, it is helpful to contextualize concepts and statistics with respect to a specific strategy. The next few chapters will focus on a *short strangle*, an options strategy consisting of a short out-of-the-money (OTM) call and a short OTM put:

- A short OTM call (the right to buy an asset at a certain price) has a bearish directional assumption. The seller profits when the underlying price stays below the specified strike price.
- A short OTM put (the right to sell an asset at a certain price) has a bullish directional assumption. The seller profits when the underlying price stays above the specified strike price.

These two contracts combine to form a strangle. This is an example of an *undefined risk* strategy, where the loss is theoretically unlimited. The short call has undefined risk because stock prices can increase indefinitely, meaning the potential loss to the upside is unknown. Though short puts technically cannot lose more than 100 times the strike price, this potential loss is large enough that they are also considered undefined risk. Defined risk strategies, where the maximum loss is limited by the construction of the trade, have pros and cons that will be discussed in [Chapter 5](#). For simplicity, the strangle is used to formulate most examples in this book.

Strangles have a neutral directional assumption for the contract seller, meaning it is typically profitable when the price of the underlying stays within the range defined by the short call strike and short put strike. Investors often define the strikes of a strangle according to the expected range of the underlying price (or some multiple of the expected range) over the contract duration. The one standard deviation expected range can be approximated with the current implied volatility of the underlying, as shown in [Chapter 2](#).



**Figure 3.1** The price of SPY in the last five months of 2019. Included is the 45-day expected move cone calculated from the IV of SPY in December 2019. The edges of the cones are labeled according to appropriate strikes for an example strangle.

**Figure 3.1** shows that SPY was priced at roughly \$315 around December 2019, when the current IV for SPY was 12% (corresponding to a VIX level of 12). This means the price for SPY was forecasted to move between – 4.2% and +4.2% over the next 45 days with a 68% certainty. This is equivalent to a 45-day forecast of the price of SPY staying between \$302 and \$328 approximately. A contract with a strike price corresponding to the

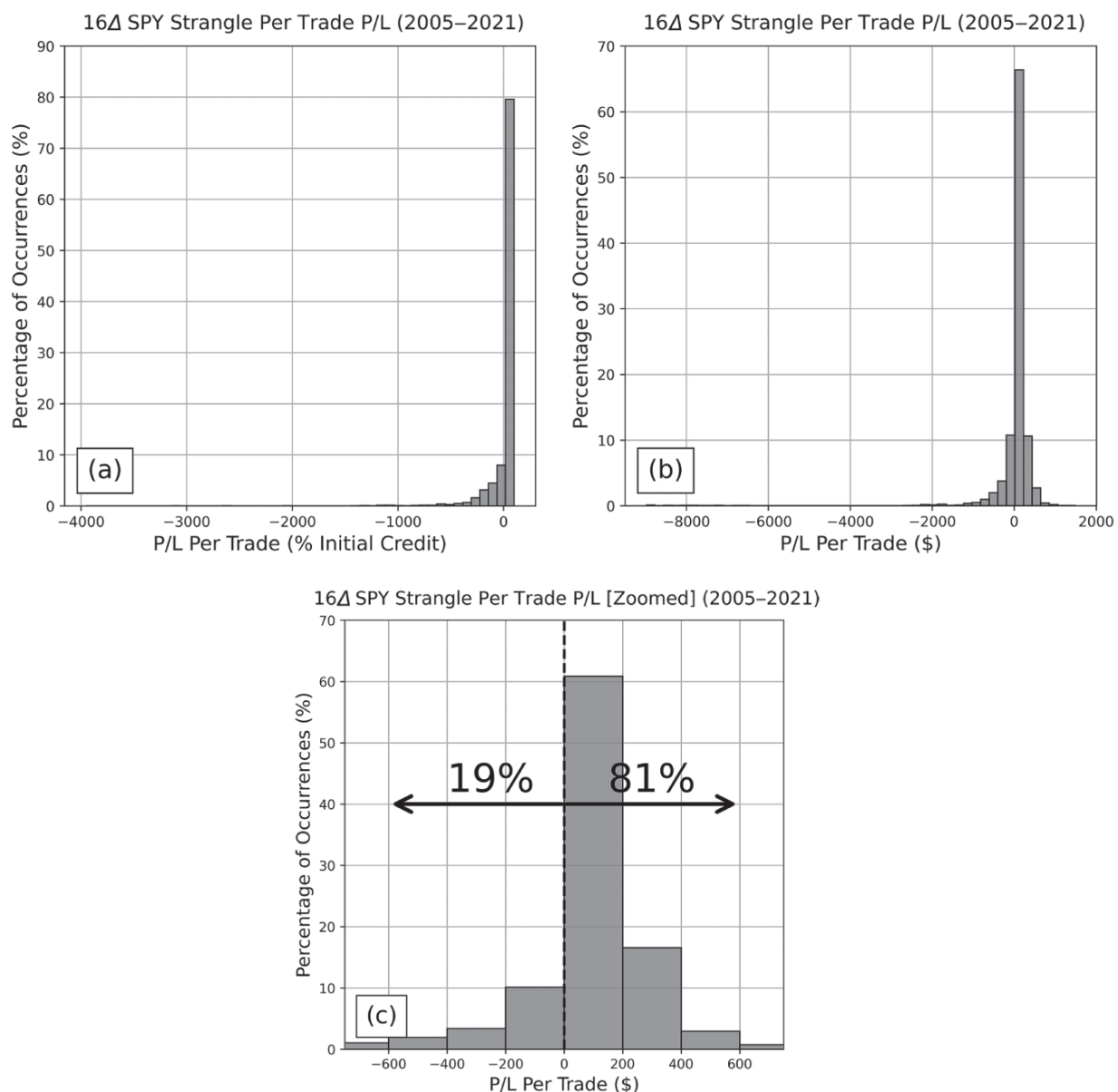
$1\sigma$  expected move range is approximately a 16  $\Delta$  contract. In this scenario, a 45 days to expiration (DTE) short SPY call with a strike price of \$328 is a  $-16\Delta$  contract roughly, and a 45 DTE short SPY put with a strike price of \$302 is approximately a 16  $\Delta$  contract. The two positions combined form a delta-neutral position known as a 45 DTE 16  $\Delta$  SPY strangle.<sup>1</sup>

The strangle buyer and seller are making different bets:

- The strangle buyer assumes that SPY's price will move beyond expectation within the next 45 days, either to the upside or the downside. More specifically, the long strangle yields profit if the price of SPY significantly increases above \$328 or decreases below \$302 prior to expiration.
- The strangle seller profits if the position expires when the underlying price is within or near its expected range or if the position is closed when the contract is trading for a cheaper price than when it was opened (IV contraction).

Because there is a 68% chance the underlying will stay within its expected range, the short position theoretically has a 68% chance of being profitable. However, since the underlying price tends to stay in its expected range more often than theoretically predicted, this results in the probability of profit (POP) of short strangles held to expiration being much higher.

For example, consider the profit and loss (P/L) distributions for the short 45 DTE 16  $\Delta$  SPY strangle in [Figures 3.2\(a\)–\(c\)](#). These distributions were generated using historical options data and are useful for visualizing the long-term risk-reward profile and likely trade-by-trade outcomes for this type of contract. Each occurrence in the histogram corresponds to the final P/L of a short strangle held to expiration.<sup>2</sup> P/L can be represented as a raw dollar amount or as a percentage of initial credit (the fraction of option premium that the seller ultimately kept).<sup>3</sup>



**Figure 3.2** (a) Historical P/L distribution (% of initial credit) for short 45 DTE 16Δ SPY strangles, held to expiration from 2005–2021. (b) Historical P/L distribution (\$) for short 45 DTE 16Δ SPY strangles, held to expiration from 2005–2021. (c) The same distribution as in (b) but zoomed in near \$0. The percentage of occurrences on either side of \$0 have been labeled.

**Figure 3.2(c)** shows that 81% of occurrences are *positive* and only 19% are *negative*. This means this strategy has historically profited 81% of the time and only taken losses 19% of the time, significantly higher than the 68% POP that the simplified theory suggests. Over the long run, this strategy

was *profitable* and averaged a P/L of \$44 (or 28% of the initial credit) per trade. However, notice the P/L distributions for this strategy are highly skewed and carry significant tail risk. As shown in [Figure 3.2\(a\)](#), these tail losses are unlikely but could potentially amount to  $-1,000\%$  or even  $-4,000\%$  of the initial credit. In other words, if a trader receives \$100 in initial credit for selling a SPY strangle, there is a slim chance of losing upward of \$4,000 on that trade according to historical behavior. This is the trade-off for the high POPs of short premium strategies.

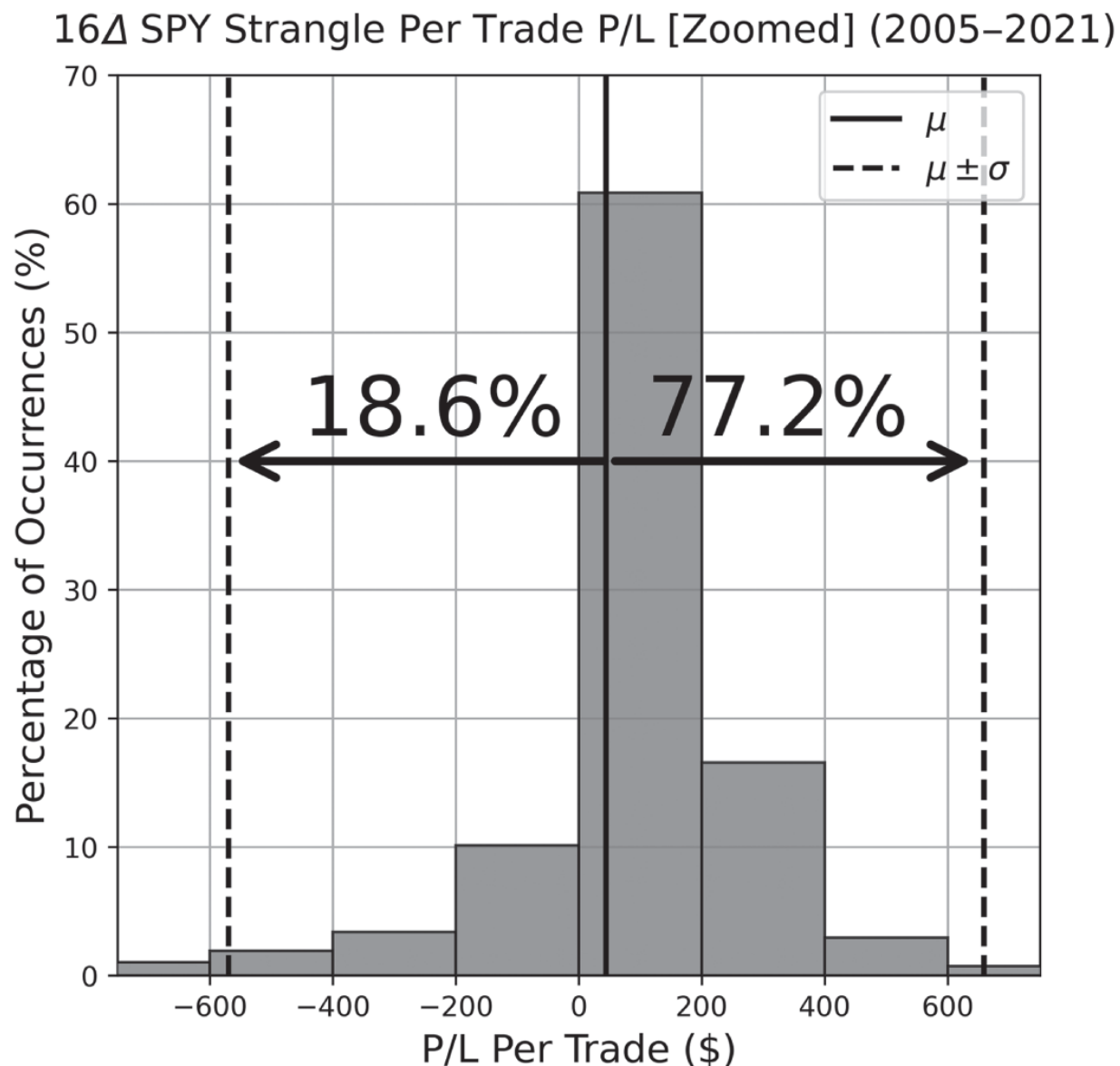
The possibility of outlier losses should not be surprising because placing a short premium trade is betting against large, unexpected price swings. For a relatively stable asset like SPY, these types of swings rarely happen. When they do, things can fly off the handle rapidly, such as during the 2008 recession or 2020 sell-off. Consequently, the most important goals for a short premium trader are to profit consistently enough to cover moderate, more likely losses and to construct a portfolio that can survive those unlikely extreme losses.

## Background: A Note on Quantifying Option Risk

Approximating the historical risk of a stock or exchange-traded fund (ETF) is relatively straightforward. Equity log returns distributions are fairly symmetric and resemble a normal distribution, thus justifying that standard deviation of returns (historical volatility) be used to approximate historical risk. However, a short option P/L distribution is highly skewed and subject to substantial outlier risk. Due to this more complex risk profile, using option P/L standard deviation as a lone proxy for risk *significantly* misrepresents the true risk of the strategy. Therefore, the following metrics will be used to more thoroughly discuss the risk of short options: standard deviation of P/L, skew, and conditional value at risk (CVaR).<sup>4</sup>

**The standard deviation of P/L** encompasses the range that the *majority* of endings P/Ls fall within for a given strategy historically. The standard deviation for financial strategies is commonly interpreted relative to the normal distribution, where one standard deviation accounts for 34% of the distribution on either side of the mean. For options P/L distributions,

however, the one standard deviation of P/L typically accounts for more than 68% of the total occurrences and the density of occurrences is not symmetric about the mean. Again, consider the P/L distribution for the short 45 DTE 16  $\Delta$  SPY strangle.



**Figure 3.3** Historical P/L distribution (\$) for 45 DTE 16  $\Delta$  SPY strangles, held to expiration from 2005–2021. The distribution has been zoomed in near the mean (solid line), and the percentage of occurrences within  $\pm 1\sigma$  of the mean has been labeled.

For 45 DTE 16  $\Delta$  SPY strangles from 2005–2021, the average P/L was \$44, and the standard deviation of P/L was \$614. As shown in [Figure 3.3](#),

the one standard deviation range accounts for nearly 96% of all occurrences, significantly higher than the  $\pm 1\sigma$  range for the normal distribution. Additionally, because the distribution is highly asymmetric, the P/Ls in the  $-1\sigma$  range are less likely than the P/Ls in the  $+1\sigma$  range. Due to these factors, the interpretation of standard deviation as a measure of risk must be adjusted. Standard deviation *overestimates* the magnitude of the most likely losses (e.g., a \$500 loss is unlikely, but the standard deviation range does not clarify that) and does not account negative tail risk. It does yield a range for the *most likely* profits and losses on a trade-by-trade basis for a given strategy. Therefore, traders can generally form more reliable P/L expectations for strategies with a lower P/L standard deviation.

**Skew and CVaR** are used to estimate the historical tail risk of a strategy. As covered in [Chapter 1](#), skew is a measure of the asymmetry of a distribution. Strategies with a larger magnitude of negative skew in their P/L distribution have more historical outlier loss exposure. CVaR gives an estimate of the potential loss of a position over a given time frame at a specific likelihood level based on historical behavior. CVaR can be used to approximate the magnitude of an expected worst-case loss and contextualize skew. For example, consider the two example short strangles outlined in [Table 3.1](#).

**Table 3.1** Two example short strangles. For Strangle A, CVaR estimates losing at least \$200 at most 5% of the time. In this example, the time frame for this loss has not been specified, but one may assume the time frame is identical for both strategies.

| Risk Factors | Strangle A | Strangle B |
|--------------|------------|------------|
| Skew         | -5.0       | -1.0       |
| CVaR (5%)    | -\$200     | -\$2,000   |

Strangle A has a larger magnitude of negative skew, indicating that this strategy is more susceptible to tail risk and outlier losses compared to Strangle B. However, there is 10 times more capital at risk in an extreme loss scenario for Strangle B compared to Strangle A perhaps because the underlying for Strangle B is more expensive. Generally speaking, strategies with less skew are preferable because those strategies are less susceptible to

large, unpredictable losses and perform more consistently. However, the optimal trade ultimately depends on the acceptable amount of per-trade capital at risk according to the trader's personal preferences.

Also note it is difficult to accurately model outlier loss events because they happen rarely. P/L distributions can give an *idea* of the magnitude of extreme losses, but these statistics are averaged over a broad range of market conditions and volatility environments. They are not necessarily representative of outlier risk at the present time. Buying power reduction (BPR), which will be covered in the next chapter, yields an estimate for the worst-case loss of a trade according to current market conditions. Similar to implied volatility, BPR is a forward-looking metric designed to encompass the most likely scope of losses for an undefined risk position.

## Trading in High IV

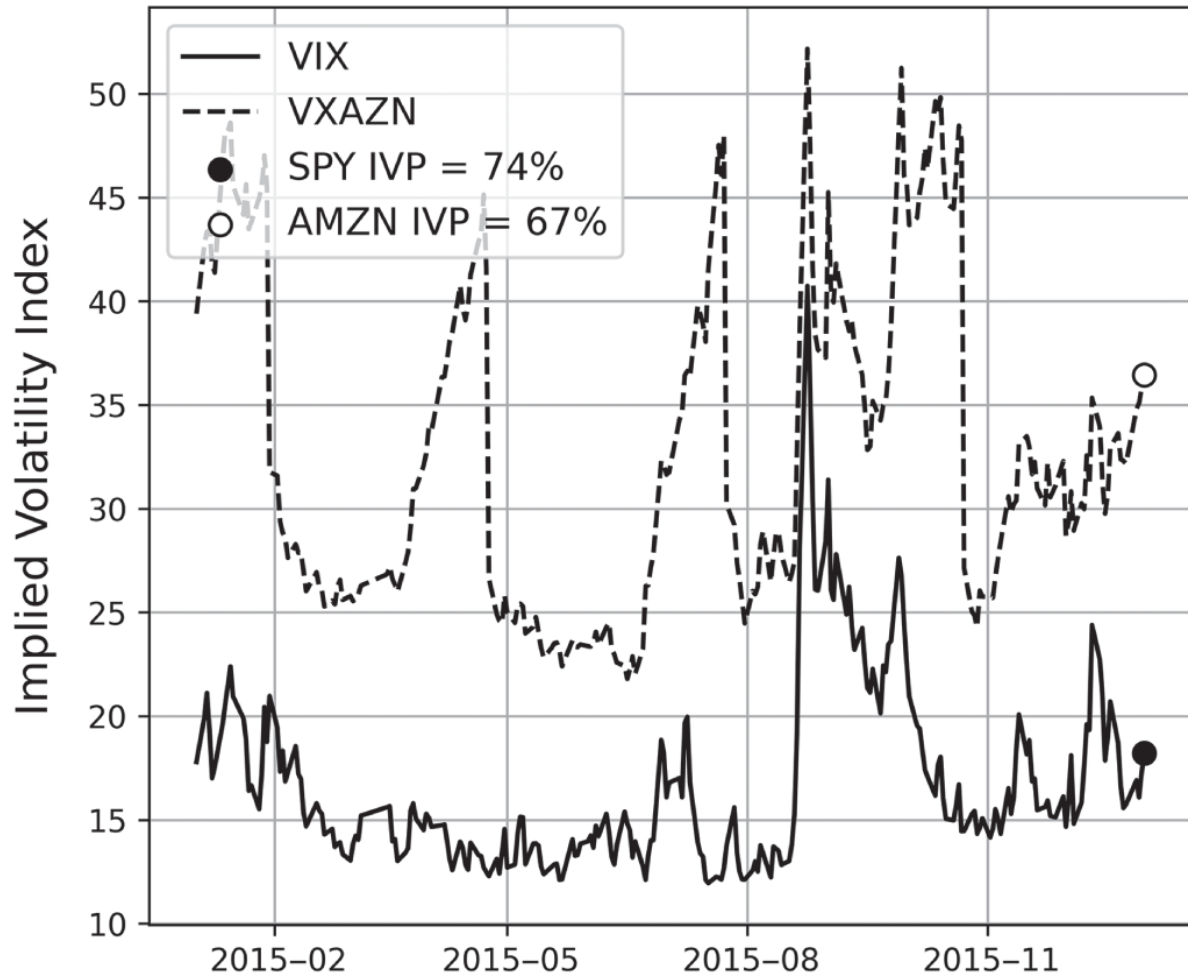
Selling premium once IV is elevated comes with several advantages. Before that discussion, there are subtleties to note when evaluating “how high” the IV of an asset is. Contextualizing the current IV for an asset like SPY is somewhat straightforward because it has a well-known and widely available IV index. The VIX has historically ranged from approximately 10 to 90, has an average of roughly 18, is typically below 20, and rarely surpasses 40. Therefore, a trader can intuitively interpret a level of 15 as fairly low and a level of 35 as fairly high relative to the long-term behavior of the VIX. But how do traders contextualize the current IV relative to a shorter timescale, such as the last year? And how do traders contextualize the current IV for a less popular IV index with a totally different risk profile? For example, is 35 high for VXAZN, the IV index for AMZN?

One way to gauge the degree of IV elevation with respect to some timescale is by converting raw implied volatility into a relative measure such as IV percentile (IVP). IVP is the percentage of days in the past year where the IV was *below* the current IV level, calculated with the following equation. Note that 252 is the number of trading days in a year.

$$\text{IVP} = \frac{\text{Number of days in past year with IV below current IV}}{252} \quad (3.1)$$

IVP ranges from 0% to 100%, with a higher number indicating a higher relative IV. This metric normalizes raw IV to put the current level in context, and unlike raw IV, it is comparable between assets. For example, consider the raw IV indexes and the corresponding IVP values for SPY and AMZN shown in [Figure 3.4](#).

## IV Index and IV Percentile Comparison (2015–2016)



**Figure 3.4** The VIX (solid) and VXAZN (dashed) from 2015–2016. Labeled are the IVP values for each index at the end of 2015. When the VIX was roughly 18 SPY had an IVP of 74%, and VXAZN was roughly 36 AMZN had an IVP of 67%.

At the end of 2015, the VIX was near its long-term average of 18 and would have been considered low. However, market IV was below average for the majority of 2015, and a VIX level of 18 was higher than nearly 74% of occurrences from the previous year. A SPY IVP of 74% indicates that IV is fairly elevated relative to the recent market conditions, suggesting that volatility may contract following this expansion period. Comparatively, the volatility index for AMZN at the end of 2015 was 37. This is significantly higher than the VIX at the time but is actually *less elevated* relative to its volatility history from the past year according to the AMZN IVP of 67%.

SPY and AMZN have dramatically different volatility profiles, with VXAZN frequently exceeding 35 and the VIX rarely doing so. This makes raw IV a poor metric for comparing relative volatility and a metric like IVP necessary.

Another commonly used relative volatility metric is IV rank (IVR), which compares the current IV level to the historical implied volatility range for that underlying. It is calculated according to the following formula:

$$\text{IVR} = \frac{\text{Current IV} - \text{Min. IV over past year}}{\text{Max. IV over past year} - \text{Min. IV over past year}} \quad (3.2)$$

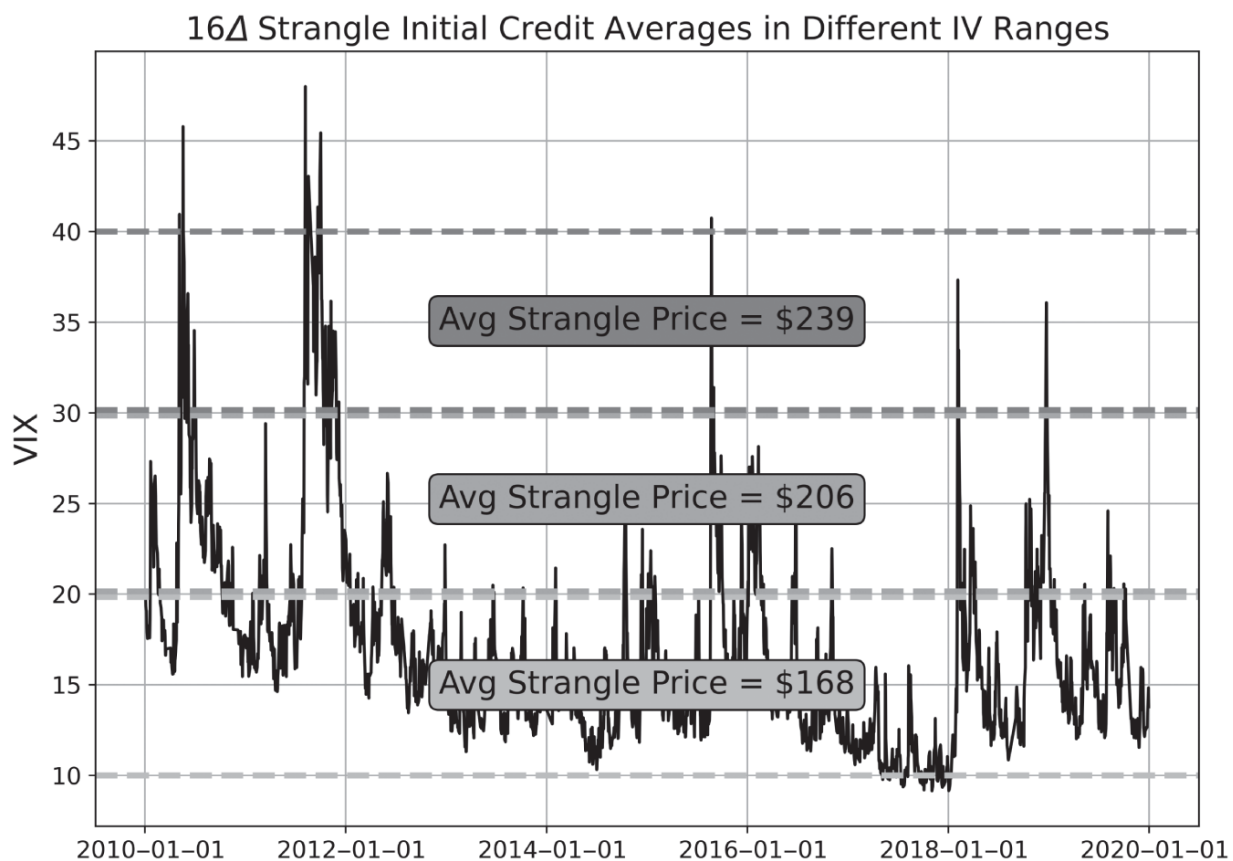
Similar to IVP, IVR normalizes raw IV on a 0% to 100% scale and is comparable between assets. IVR gives a better direct metric for evaluating the price of an option compared to IVP. However, IVP is more robust than IVR because IVR is more sensitive to outlier moves and prone to skew.

Both metrics are suitable for practical decision making because they assist traders with evaluating current volatility levels and selecting a suitable strategy/underlying for those conditions. They are also useful for identifying suitable, high IV underlyings for a portfolio because most assets do not have well-known volatility indices. However, both metrics are fairly unstable, sensitive to timescale, and can be skewed by prolonged outlier events such as sell-offs. Raw IV, assuming that the characteristics of the volatility profile are well understood, is generally a more stable and reliable metric for analyzing long-term trends. Because most studies throughout this book use SPY as a baseline underlying and span several years, raw IV will be used rather than a relative metric.

As previously mentioned, trading short premium when IV is elevated comes with the added benefits of higher credits and more profit potential for sellers. This is shown in [Figure 3.5](#), which includes average credits for 16  SPY strangles from 2010–2020 in different volatility environments.

Trading short premium in elevated IV is an effective way to capitalize on higher premium prices and the increased likelihood of a significant volatility contraction. Trading when credits are higher also means common losses tend to be larger (as a dollar amount), but the exposure to outlier risk

actually tends to be *lower* when IV is elevated compared to when it's closer to equilibrium. This may seem counterintuitive: If market uncertainty is elevated and there is higher perceived risk, wouldn't short premium strategies carry more outlier risk? Although moves in the underlying tend to be more dramatic when IV is high, the expected range adjusts to account for the new volatility almost immediately, which in many cases reduces the risk of an outlier loss. To understand this, consider [Figures 3.6\(a\) and \(b\)](#), showing extreme losses for 16  $\Delta$  SPY strangles from 2005–2021, with an emphasis on the 2008 recession.

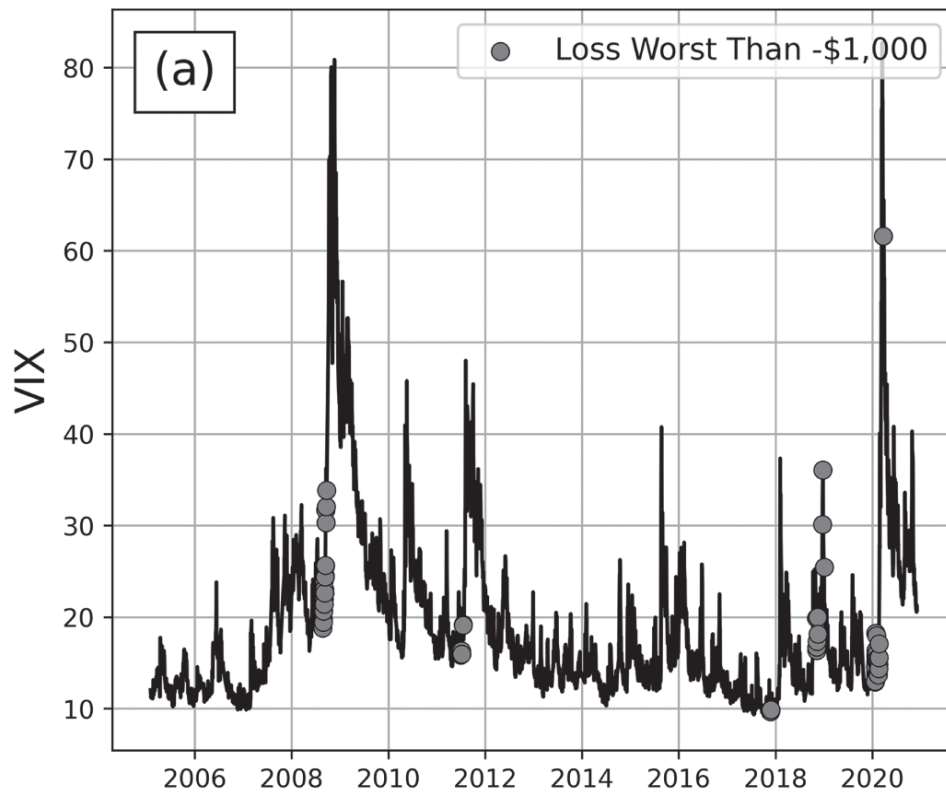


**Figure 3.5** SPY IV from 2010–2020. The average prices for 45 DTE 16  $\Delta$  SPY strangles are labeled at different VIX levels: 10–20, 20–30, and 30–40. When the VIX was between 30 and 40, the average initial credit per one lot for the 16  $\Delta$  SPY strangle was roughly 42% higher than when the VIX was between 10 and 20.

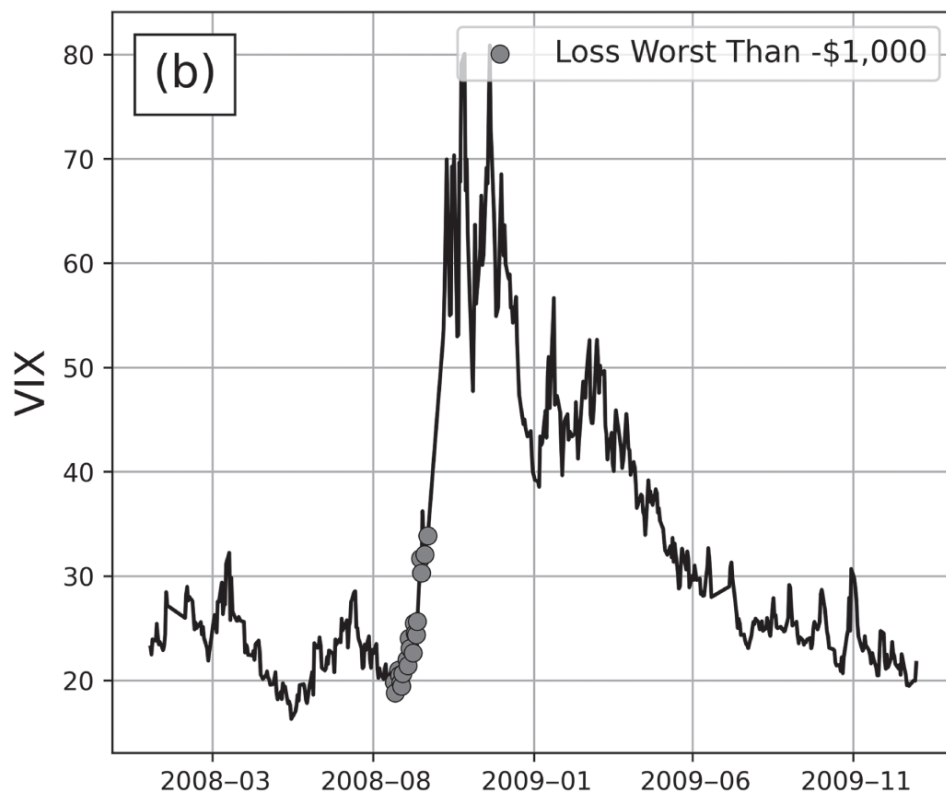
A short 16  $\Delta$  SPY strangle rarely incurs a loss over \$1,000. From 2005–2021, this occurred less than 1% of the time. However, 84% of these losses occurred when the VIX was below 25. During the initial IV expansion of

the 2008 recession (late August to early October), strangles incurred these large losses approximately 56% of the time. Notice in [Figure 3.6](#) that these extreme losses were confined to the initial IV expansion (when the VIX increased from roughly 20 to 35). This is because the market was not anticipating the large downside moves of the recession, as reflected by the VIX being near its long-term average of 18. Because these large swings happened when the expected move range was tight, the historical volatility of the market well exceeded its expected range, and long strangles were highly profitable. Once market uncertainty adjusted to the new conditions and initial credits and expected ranges increased to reflect the perceived risks, the outlier losses for short strangles diminished.

16Δ SPY Strangle Large Losses (2005–2021)



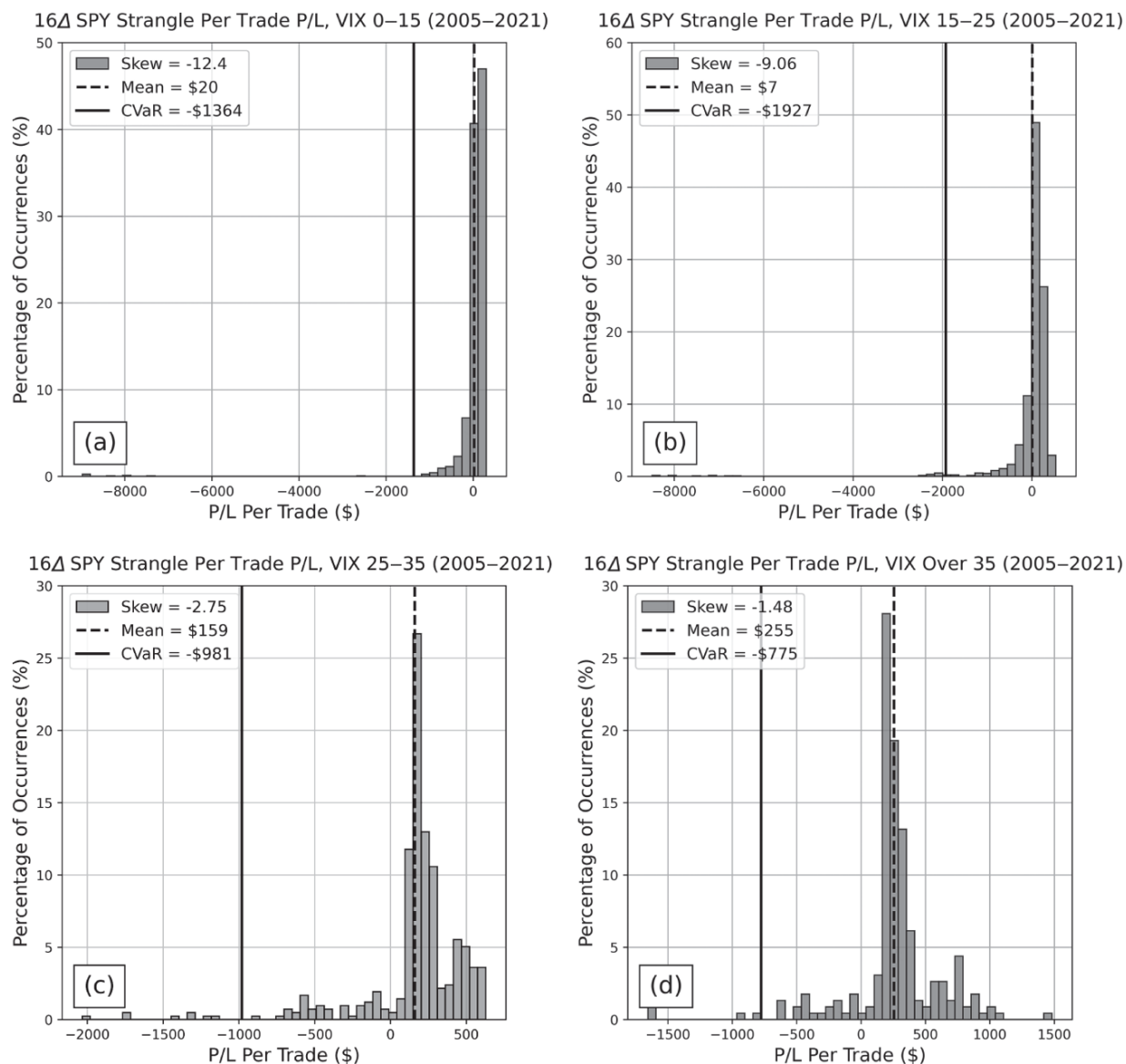
16Δ SPY Strangle Large Losses (2008 Recession)



**Figure 3.6** (a) SPY IV from 2005–2021. Labeled are the extreme losses for 45 DTE 16  $\Delta$  SPY strangles held to expiration, meaning losses that are worse than \$1,000. (b) The same figure as shown in (a) but zoomed in to 2008–2010, during the 2008 recession.

These unexpected periods of high market volatility are the primary source of extreme loss for short premium positions. These events typically happen when there are large price swings in the underlying and the expected move range is tight (low IV). These extreme expansion events are rare, and trading short premium once IV is elevated tends to reduce this type of exposure. Another way to demonstrate this concept is to consider the amount of skew in the P/L distribution of the 16  $\Delta$  SPY strangle at different IV levels.

**Figures 3.7(a)–(d)** illustrate that strangle P/L distributions have less negative skew and smaller tail losses as IV increases. This means that, historically, the exposure to negative tail risk was much higher when the VIX was closer to the lower end of its range compared to when the VIX already expanded. The P/L distribution becomes more symmetric as IV increases, indicated by the decreasing magnitude of negative skew. This means that higher IV conditions facilitate more dependable profit and loss expectations than lower IV conditions. As an important note, observe that there are significantly fewer occurrences when the VIX was over 35 (a few hundred occurrences) compared to when the VIX was between 0 and 25 (thousands of occurrences). This brings us to the next point to consider: How often should one trade?



**Figure 3.7** Historical P/L distributions for 45 DTE 16Δ SPY strangles, held to expiration from 2005–2021: (a) Occurrences when the VIX is between 0 and 15 (1,603 occurrences total), (b) Occurrences when the VIX is between 15 and 25 (1,506 occurrences total). (c) Occurrences when the VIX is between 25 and 35 (416 occurrences total). (d) Occurrences when the VIX is above 35 (228 occurrences total).

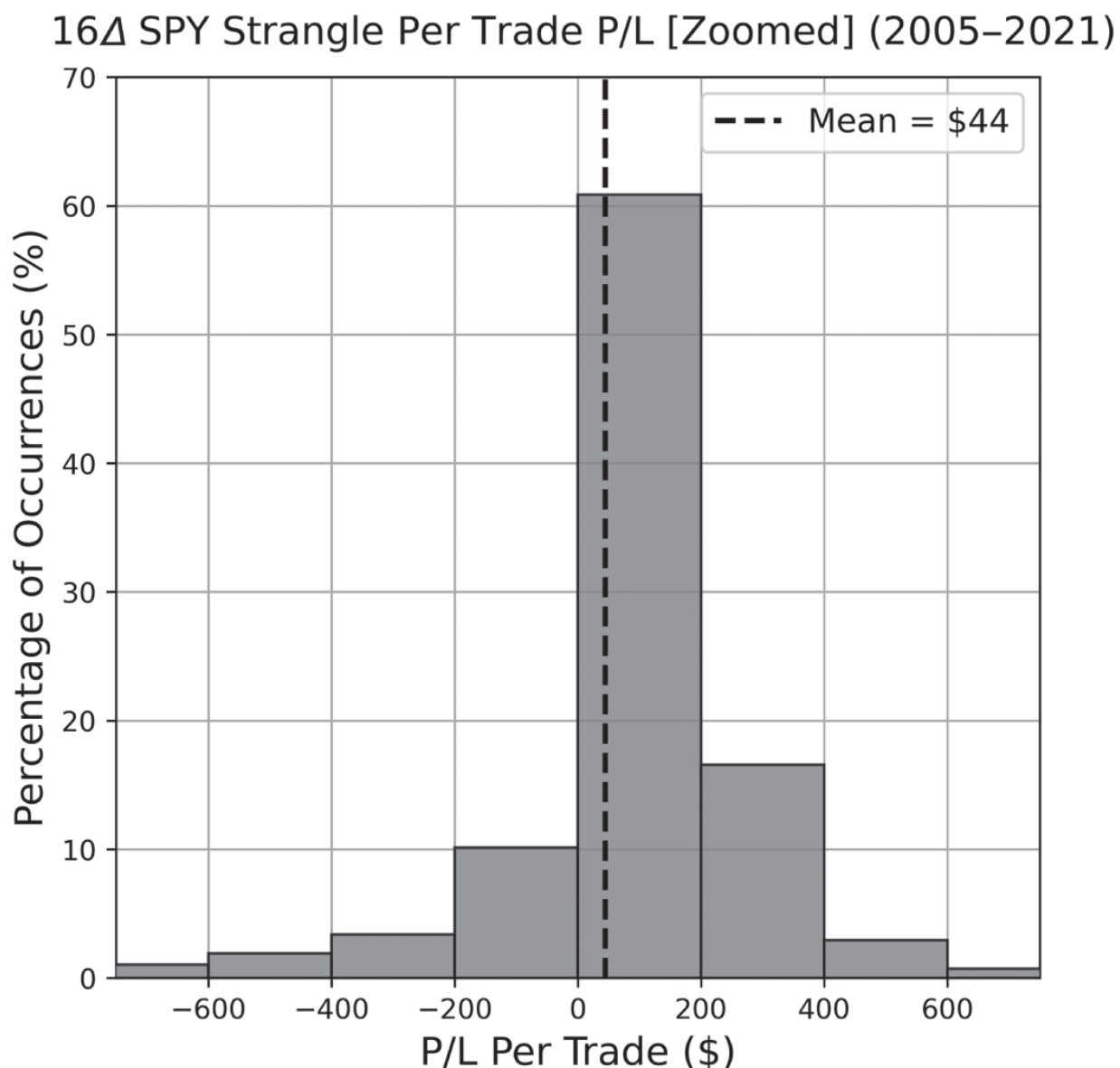
## Number of Occurrences

Table games at a casino typically have maximum bet sizes. The house has the statistical edge for every game in the casino, but the house will not

necessarily profit from that edge unless patrons bet *often*. In blackjack, the house has an edge of 0.5% if the player's strategy is statistically optimized. So, if gamblers wager \$100,000 on blackjack throughout the night, they should lose approximately \$500 to the house after a sufficiently large number of hands. If the opponent plays 10 hands at \$10,000 per hand, they may win eight hands, three hands, or even all 10 hands. In this case, the variance of potential outcomes is fairly large, and the casino may have to pay fairly large payouts. However, if the opponent plays 1,000 hands at \$100 per hand, it is more likely the player's loss will amount to the expected \$500.

By capping bet sizes, the casino aims to increase the number of occurrences from a single gambler so the house is more likely to reach long-run averages for each game, a consequence of the law of large numbers and the central limit theorem. When a *small* number of events is randomly sampled from a probability distribution repeatedly and the averages of those samples are compared, the variance of those averages tends to be quite large. But as the number of occurrences increases, the variance of the averages decreases and the sampled means converge to the expected value of the distribution.<sup>5</sup>

Just as the casino aims to realize the long-term edge of table games by capping bet sizes and increasing the number of plays, short premium traders should make many small trades to maximize their chances of realizing the positive long-run expected averages of short premium strategies. For an example of why this is crucial, refer again to the P/L distribution of the 16  SPY strangle.

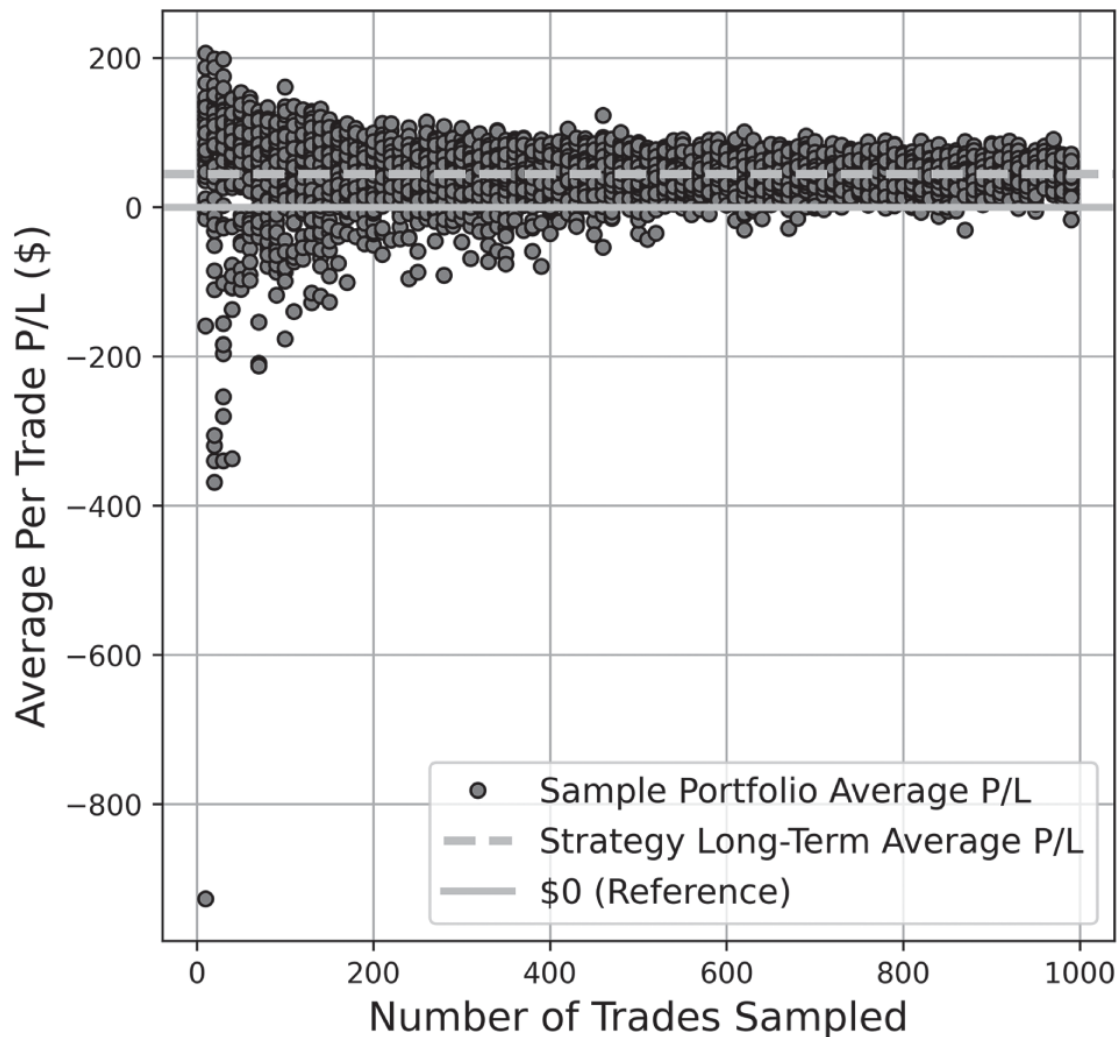


**Figure 3.8** Historical P/L distribution for 45 DTE 16Δ SPY strangles, held to expiration from 2005–2021. The dotted line is the long-term average P/L of this strategy.

This strategy, shown in [Figure 3.8](#), has an average P/L per trade of roughly \$44 and a POP of 81%. However, these long-term averages were calculated using roughly 3,750 trades. Calculating averages with a large pool of data provides the least amount of statistical error but does not model the occurrences retail traders can realistically achieve. What P/L would short premium traders have averaged if they only placed 10 trades from 2005 to 2021? 100 trades? 500 trades? [Figure 3.9](#) shows a plot of average P/Ls for a

collection of sample portfolios, each with a different number of trades randomly selected from the P/L distribution of the 16  $\Delta$  SPY strangle.

### Portfolios of 16 $\Delta$ SPY Strangles (Sampled from 2005–2021)



**Figure 3.9** P/L averages for portfolios with  $N$  number of trades, randomly sampled from the historical P/L distribution for 45 DTE 16  $\Delta$  SPY strangles, held to expiration from 2005–2021. The variance among these portfolio averages is very large when a small number of trades are sampled. As more trades are sampled, the averages converge to the long-term average P/L of this strategy.

As you can see, when a small number of trades is sampled, 10 for example, the average P/L ranges from roughly  $-\$900$  to  $\$200$ . This means that if two traders randomly traded 10 short strangles from 2005 to 2021, one trader

may have profited by \$2,000, and the other may have lost \$9,000. As the number of occurrences increases, the variance of P/L averages among these sample portfolios decreases, and the averages converge toward the long-run expected value of this strategy. In other words, if two traders randomly traded 1,000 short strangles from 2005 to 2021, it would be fairly likely for both to average a P/L near \$44 per trade, the historical long-term average P/L of this strategy.

Number of occurrences is a critical factor in achieving long-term averages, and the minimum number of occurrences needed varies with the specific strategy's standard deviation of P/L. For practical purposes, a minimum of roughly 200 occurrences is necessary to reach long-run averages, and more is better. This puts short premium traders in a bit of a predicament because, although trading short premium in high IV is ideal, high IV environments are very uncommon as shown in [Table 3.2](#).

**[Table 3.2](#)** How often the VIX fell in a given range from 2005–2021.

| VIX Data (2005–2021) |                  |
|----------------------|------------------|
| VIX Range            | % of Occurrences |
| 0–15                 | 43%              |
| 15–25                | 40%              |
| 25–35                | 11%              |
| 35+                  | 6%               |

The VIX is at the low end of its range 43% of the time and below 18.5, its long-term average, the majority of the time. From 2005–2021, the VIX was only above 35 roughly 6% of the time, which does not leave much opportunity for trading short premium in very high IV. To optimize the likelihood of reaching the favorable long-term expected values of this strategy, it is clearly necessary to trade in non-ideal, low volatility conditions. The next section covers how to trade in all market conditions while mitigating the outlier risk in low volatility environments, specifically by maintaining small position sizes and limiting the capital exposed to outlier losses.

## Portfolio Allocation and Position Sizing

In practice, short premium traders must strike a balance between being exposed to large losses and reaching a sufficient number of occurrences. Trading in high IV tends to carry less exposure to outlier risk compared to trading in low IV, but trading in low IV is still profitable on average. Unlike long stocks, which are only profitable during bullish conditions, short options may be profitable in bullish, bearish, or neutral conditions and spanning all volatility environments. For the 16  $\Delta$  SPY strangle from 2005–2021, for example, the majority of occurrences were profitable in all IV ranges. (See [Table 3.3](#).)

**Table 3.3** The POPs and average P/Ls in different IV ranges for 45 DTE 16  $\Delta$  SPY strangles, held to expiration from 2005–2021.

| 16 $\Delta$ SPY Strangle Data (2005–2021) |     |             |
|-------------------------------------------|-----|-------------|
| VIX Range                                 | POP | Average P/L |
| 0–15                                      | 82% | \$20        |
| 15–25                                     | 78% | \$7         |
| 25–35                                     | 86% | \$159       |
| 35+                                       | 89% | \$255       |

By trading short options strategies in all IV environments, profits accumulate more consistently, and the minimum number of occurrences is more achievable. To manage exposure to outlier risk throughout these environments, it's *essential* to keep position sizes small and limit the total amount of portfolio capital allocated to short premium positions, which can be scaled according to the current outlier risk. The percentage of portfolio capital allocated to short premium strategies should generally range from 25% to 50%, with the remaining capital either kept in cash or a low-risk passive investment.<sup>6</sup> This is because allocating less than 25% severely limits upside growth, while allocating more than 50% may not leave enough capital for a portfolio to recover from an outlier loss event. Because the exposure to outlier risk tends to be higher when IV is low, scaling allocation down in low IV protects portfolio capital from the tail exposure of unexpected market volatility. Once IV increases, scaling short premium capital allocation up increases the potential to profit from higher credits, larger profits, and reduced outlier risk.

**Table 3.4** Guidelines for allocating portfolio capital according to market IV.

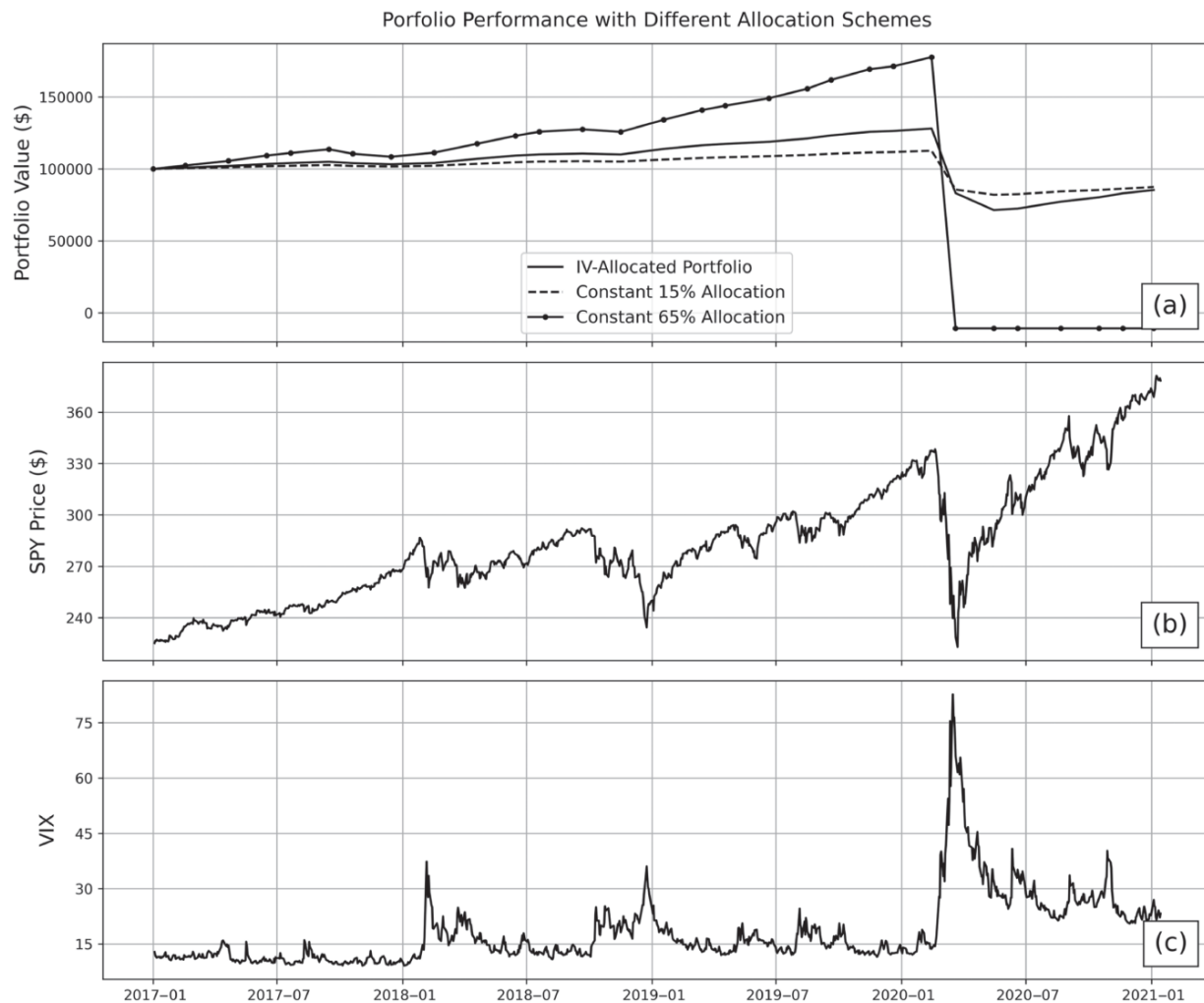
| VIX Range | Max Portfolio Allocation |
|-----------|--------------------------|
| 0–15      | 25%                      |
| 15–20     | 30%                      |
| 20–30     | 35%                      |
| 30–40     | 40%                      |
| 40+       | 50%                      |

A portfolio should not be overly concentrated in short options strategies for the given market conditions, and the capital allocated to short premium should *also* not be overly concentrated in a single position. An appropriately sized position should not occupy more than 5% to 7% of portfolio capital. The exact percentage varies depending on the POP of the strategies used, and this will be covered in more detail in [Chapter 8](#).

To understand why it's crucial to limit capital exposure and beneficial to scale portfolio allocation according to IV, look at a worst-case scenario: the 2020 sell-off. The 2020 sell-off produced historic losses for short premium positions. From late February to late March 2020, the price of SPY crashed by roughly 34%. For 45 DTE 16  $\Delta$  SPY strangles, the most extreme losses recorded for this position occurred throughout this time. A 16  $\Delta$  SPY strangle opening on February 14, 2020, and expiring on March 20, 2020, had a P/L per one lot of roughly  $-\$8,974$ , the worst recorded loss in 16 years for this type of contract. If traders allocated different percentages of a \$100,000 portfolio to short SPY strangles beginning with this worst-case loss, how would those portfolios perform in regular market conditions compared to highly volatile conditions like the 2020 sell-off? Compare three portfolio allocation strategies: allocation by IV guidelines (25–50%), a more conservative allocation (constant 15%), and a more aggressive allocation (constant 65%).<sup>7</sup>

Unsurprisingly, the portfolios perform markedly differently in regular conditions compared to the 2020 sell-off. From 2017 to February of 2020, the aggressive portfolio dramatically outperformed the conservative and IV-allocated portfolios. Throughout this three-year period, the conservative portfolio grew by 13% and the IV-allocated portfolio by 28%, and the

aggressive portfolio increased by 78%. Comparatively, from 2017–2020, SPY grew by 50%. This means that a \$100,000 portfolio fully allocated to SPY shares would have outperformed the conservative and IV-allocated portfolios but underperformed the aggressive portfolio, though it would have required significantly more capital than any of them.



**Figure 3.10** (a) Performances from 2017 to 2021, through the 2020 sell-off. Each portfolio has different amounts of capital allocated to approximately 45 DTE 16  $\Delta$  SPY strangles that are closed at expiration and reopened at the beginning of the expiration cycle. The portfolios are (a) IV-allocated (solid), conservative (dashed), and aggressive (hashed). (b) SPY price from 2017 to 2021. (c) VIX throughout the same time frame.

Upon the onset of the highly volatile market conditions of 2020, the highly exposed aggressive portfolio was immediately wiped out. The conservative

and IV-allocated portfolios were also impacted by significant losses and declined by 35% and 24%, respectively, from February to March 2020. In all the previous scenarios, each portfolio experienced some degree of loss during the extreme market conditions of the 2020 sell-off. The important thing to note is that portfolios with less capital exposure and position concentration ultimately had the capital to recover following these losses. Following the 2020 sell-off, the conservative portfolio recovered by 7% and the IV-allocated by 20% because this portfolio was able to capitalize on the high IV and higher credits of the sell-off recovery.

For profit goals to be reached *consistently*, it's crucial to construct a portfolio that is robust in every type of market. A highly exposed portfolio takes extraordinary profits in more regular market conditions, but there is a high risk of going under in the rare event of a sell-off or major correction. A more conservative portfolio is well suited for extreme market conditions, but upside profits are limited the majority of the time. Comparatively, scaling capital allocation according to market IV is an effective way to capitalize on higher profits when IV is high, protect capital from outlier losses when IV is low, and achieve reasonable growth with lower capital requirements than purchasing equities directly. More importantly, limiting capital exposure and maintaining appropriate position sizes are arguably the most effective ways to minimize the impact from extreme events. These concepts will be explored in more detail in [Chapter 7](#).

## **Active Management and Efficient Capital Allocation**

Up until now, this book has discussed option risk and profitability for contracts held to expiration. However, short premium traders can also close, or manage, their positions early by purchasing long options with the same underlying, strike, and date of expiration. This can often be profitable as a result of partial theta decay and IV contractions, and it also tends to reduce P/L variability per trade. Options tend to have more P/L fluctuations in the second half of the contract duration compared to the first half, a result of increasing gamma risk. Gamma, as discussed in earlier chapters, is a measure of how sensitive a contract's delta is to changes in the price of the underlying. Gamma increases for near-the-money options as expiration

approaches, meaning that delta (and, therefore, the price sensitivity of the option) becomes more unstable in response to moves in the underlying toward the end of the contract.

Managing short positions actively, such as closing a trade prior to expiration and redeploying capital to new positions, is one way to reduce the P/L swings throughout the trade duration, as well as the per-trade loss potential and ending P/L standard deviation. Early management strategies will not necessarily reduce risk in the long term because the cumulative losses of many shorter-term trades may exceed the single loss of a longer-term trade, but they do make per-trade loss potentials more reasonable. This strategy effectively allows traders to assess the viability of a trade before P/L swings become more extreme and assess whether it is an efficient use of portfolio capital to remain in the trade. Compare how the P/Ls of 45 DTE 16  $\Delta$  SPY strangles are distributed when the contracts are held to expiration versus managed around halfway to expiration (21 DTE).

**Table 3.5** Comparison of management strategies for 45 DTE 16  $\Delta$  SPY strangles from 2005–2021 that are held to expiration and managed early. Statistics include POP, average P/L, standard deviation of P/L, and CVaR at the 5% likelihood level.

| <b>16 <math>\Delta</math> SPY Strangle Statistics (2005–2021)</b> |                           |                          |
|-------------------------------------------------------------------|---------------------------|--------------------------|
| <b>Statistics</b>                                                 | <b>Held to Expiration</b> | <b>Managed at 21 DTE</b> |
| POP                                                               | 81%                       | 79%                      |
| Average P/L                                                       | \$44                      | \$30                     |
| Average Daily P/L                                                 | \$1.29                    | \$1.60                   |
| Standard Deviation of P/L                                         | \$614                     | \$260                    |
| CVaR (5%)                                                         | –\$1,535                  | –\$695                   |

According to the statistics in [Table 3.5](#), strangles managed at 21 DTE carry significantly less negative tail risk and P/L standard deviation on a trade-by-trade basis than strangles held to expiration. Additionally, although early-managed contracts collect less on average per trade, they actually average *more* profit on a daily basis and allow for more occurrences due to the shorter duration.

Managing trades early has several benefits, most of which will be covered in [Chapter 6](#). Much of this decision depends on the acceptable amount of capital to risk on a single trade and whether it is an efficient use of capital to remain in the existing trade. Notice from this example that managed trades take 24 days (21 days remaining on a 45-day duration trade corresponds to an elapsed duration of 24 days) to profit \$30 on average and held contracts 45 days to make \$44 on average. Trades may accumulate the majority of their profit potential well before expiration, depending on the market and staying in the position for the remainder of the duration may limit upside potential. Closing trades prior to expiration and redeploying capital to a new position in the same underlying is an effective method for increasing the number of occurrences in a given time frame. Redeploying that capital to a position in a different underlying with more favorable characteristics (such as higher IV) can be a more efficient use of capital and can offer elements of risk reduction in certain situations. Taking an active approach to investing and trade management provides more control over portfolio capital allocation and the flexibility to modify trades given new information.

## Takeaways

1. Compared to long premium strategies, short premium strategies yield more consistent profits and have the long-term statistical advantage. The trade-off for receiving consistent profits is exposure to large and sometimes undefined losses, which is why the most important goals of a short premium trader are to (1) profit consistently enough to cover moderate and more likely losses and (2) to construct a portfolio that can survive unlikely extreme losses.
2. Unexpected periods of high market volatility are the primary source of extreme loss for short premium positions. These events are highly unlikely but typically happen when large price swings occur in the underlying while the expected move range is tight (low IV). Trading short premium once IV is elevated is one way to consistently reduce this exposure.
3. The profitability of short options strategies depends on having a large number of occurrences to reach positive statistical averages. At

minimum, approximately 200 occurrences are needed for the average P/L of a strategy to converge to long-term profit targets and more is better.

4. Although trading short premium in high IV is ideal, high IV environments are somewhat uncommon. This means that short premium traders must strike a balance between being exposed to large losses and reaching a sufficient number of occurrences. Trading short options strategies in all IV environments accumulates profits more consistently and makes it more likely to reach the minimum number of occurrences. To manage exposure to outlier risk when adopting this strategy, it's essential to maintain small position sizes and limit the amount of capital allocated to short premium positions. This strategy can also be improved by scaling the amount of capital allocated to short premium according to the current market conditions.
5. Managing positions actively is one way to reduce P/L uncertainty on a trade-by-trade basis, use capital more efficiently, and achieve more occurrences in a given time frame. The choice of whether to close a position early and redeploy capital depends on the acceptable amount of capital to risk on a single trade and whether it is an efficient use of capital to remain in the existing trade. These concepts will be explored more in [Chapter 6](#).

## Notes

- [1](#) These are approximate strikes for the  $16\Delta$  SPY strangle calculated using the equation from [Chapter 2](#). The actual strikes for a  $16\Delta$  SPY strangle are calculated using more complex estimations for expected range, which will be touched on in the appendix.
- [2](#) It is difficult to make a one-to-one comparison between equity returns and option P/Ls because these instruments operate over different timescales. The closest option analog to an equity returns distribution is a distribution for the ending P/Ls of a particular strategy.
- [3](#) Statistics represented as a percentage of initial credit are more representative of long-term values than those represented with dollars. Equity prices drift with time, meaning the prices for their options do as

well. Normalizing P/L statistics by the initial credit makes them more robust to changes in time but also makes comparisons between strategies less intuitive. This book will often represent option statistics in dollars, but remember these statistics are averaged over fairly long time frames.

- 4 These are past-looking risk metrics. Metrics of forward-looking risk include implied volatility and buying power reduction (BPR), which will be covered in the following chapter. Forward-looking metrics are the focus of this book and more relevant in applied trading, but past-looking metrics are still included for the sake of completeness and education.
- 5 Specifically, the standard deviation of the average of  $n$  independent occurrences is  $\frac{1}{\sqrt{n}}$  times the standard deviation of a single occurrence.
- 6 More specifically, the portfolio capital being referred to here is the portfolio buying power, which we will introduce in the following chapter.
- 7 This is a highly simplified backtest and should be taken with a grain of salt. These portfolios are highly concentrated in a single position and do not incorporate any complex management strategies. Options are highly sensitive to changes in timescale, meaning that a concurrent portfolio with strangles opened on slightly different days, closed on slightly different days, or with slightly different durations may have performed quite differently than the ones shown here. These backtests show one specific outcome and serve to compare the risk of different allocation percentages in a one-to-one fashion.